

**PLATE FAILURE: BEARING AND PULL-THROUGH**

---

Part: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3230**

Title: **PLATE FAILURE: BEARING AND PULL-THROUGH**

Revision: **1.2**

Date: **04/12/2020**

---

## **1 INTRODUCTION**

### **1.1 OBJECT**

This chapter presents the method to perform the local strength analysis of a metallic plate at a mechanically fastened attachment, when the rivet is subjected to shear and tension loads, vs the potential failure modes that may occur in the plate: bearing and pull-through.

Bending induced by rivet tension in the attachment plate is treated in chapter 3240 "PLATE BASE WITH TENSIONED RIVETS". Fastened system failure in chapter 3220 "FASTENERS AND BOLTS FAILURE".

### **1.2 SCOPE AND LIMITATIONS**

Mechanically fastened joints with rivet and nut fastening systems between metallic plates:

- ✓ Bearing failure in fastened metal plate
- ✓ Pull-through failure in fastened metal plate
- ✓ Net-section failure in fastened metal plate

Provided the analysed configuration was produced using the materials standards and design rules typically followed in the aerospace industry, other failure modes, as shear-out failure, shall not occur.

Composite fastened joints are treated in other Part 3.6 of this Manual.

# PLATE FAILURE: BEARING AND PULL-THROUGH

## 2 METHOD DESCRIPTION

### 2.1 BEARING FAILURE

The familiar expression to obtain the allowable bearing load of a riveted joint in a metal plate, when using protruding fasteners, is:

$$Q_{all-br} = F_{bru} \cdot d \cdot t$$

Note this allowable strength shall be truncated by the bolt shear strength for thicknesses above certain value. Indeed, when using solid rivets, the MMPDS, in table 8.1.2.1(b), proposes some penalization factors to apply to the rivet shear allowable in order to obtain this truncation value. For practical reasons, instead, the method to be applied to get the bearing strength of a fastened joint, valid for countersunk and protruding heads, shall be that defined in the HSB 21020-01. The corresponding reserve factor is then:

$$RF_{br} = Q_{all-br} / Q$$

### 2.2 PULL-THROUGH FAILURE

The pull-out/pull-through allowable load of the joint with significant tension load in the fastener shall be:

$$T_{all-po} = 0.6 \cdot F_{su} \cdot \pi \cdot d_h \cdot t \quad \dots \text{ per Niu's book (Ref. 3) section 9.9 (A)(b)(2)}$$

### 2.3 BEARING AND PULL-THROUGH INTERACTION

Interaction law to use:

$$RF = \min \{ RF_{br} , [1 / (R_{po}^2 + R_{br}^2)^{0.5}] \}$$

... where the failure indices are defined as:

$$R_{po} = T / T_{all-po}$$

$$R_{br} = Q / (F_{bru} \cdot d \cdot t)$$

Remarks:

- ✓ The failure index for bearing is not truncated, using this interaction formula, by the bolt shear cut-off that is indeed applied in the HSB method. That truncation could lead to overly conservative results.
- ✓ The combined margin is obtained for uni-axial stresses at different cross sections, what is conservative: pull-out stress around head/nut perimeter, whereas bearing stress is calculated with the projected area of the rivet shank.
- ✓ On the other hand, the combined margin is obtained using the failure indices with the stresses at applied loads and the material ultimate stresses, not accounting for the pretension effect from applied loads to failure loads. As tension stress would not really grow proportionally to external load, this is also conservative.

### PLATE FAILURE: BEARING AND PULL-THROUGH

#### 2.4 NET-SECTION FAILURE

Even normally not the primary failure mode in riveted joints following the standard design rules, the net section strength is to be verified whenever the joint is submitted to tension or shear loads.

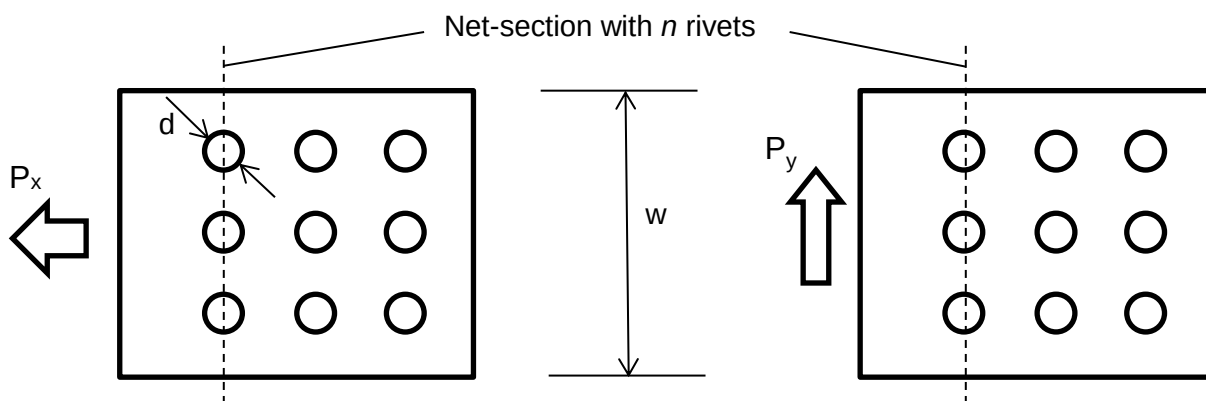


Figure 1. Critical section in tension and shear cases

Net tension stresses shall be calculated as:

$$\sigma = P_x / A_n = P_x / (w - n \cdot d) \cdot t$$

$$\tau = P_y / A_n = P_y / (w - n \cdot d) \cdot t$$

Remarks:

- ✓ With countersunk rivets the projected countersink area shall be subtracted from the net-section area.
- ✓ Expressions “as-is” are valid for rectangular rivets patterns as those in previous figures, and for many materials, also for staggered / triangular regular patterns (equilateral triangles, or  $\theta \leq 30^\circ$ ). Otherwise, some additional knock-down factors should be applied to material strengths.

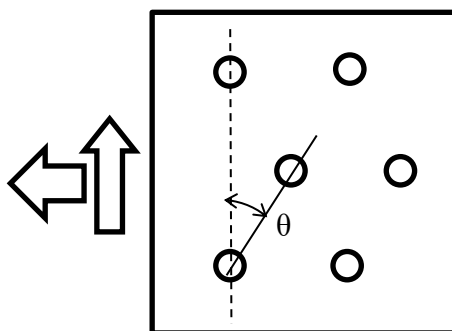


Figure 2. Staggered rivets regular pattern

The Reserve Factor shall be obtained from net-stress and plate material strengths using a Von-Mises alike interaction equation:

$$RF = 1 / [(\sigma / F_{tu})^2 + (\tau / F_{su})^2]^{0.5}$$

### PLATE FAILURE: BEARING AND PULL-THROUGH

## 3 APPENDICES

### ABBREVIATIONS AND SYMBOLS

| <i>Symbol</i> | <i>Units</i>    | <i>Definition</i>                                                                                               |
|---------------|-----------------|-----------------------------------------------------------------------------------------------------------------|
| $A_n$         | mm <sup>2</sup> | Net-section area                                                                                                |
| $d$           | mm              | Diameter of the hole                                                                                            |
| $d_h$         | mm              | Diameter of the head of the fastener in contact with the plate                                                  |
| $F_{bru}$     | MPa             | Bearing allowable stress of the plate material (taking into account edge distance)                              |
| $F_{su}$      | MPa             | Ultimate shear stress of the plate material                                                                     |
| $F_{tu}$      | MPa             | Ultimate tension stress of the plate material                                                                   |
| $n$           | -               | Number of rivets along the critical net-section (in a width $w$ ), or relative to the net-section, from context |
| $P$           | N               | Force on the plate                                                                                              |
| $Q$           | N               | Shear load                                                                                                      |
| $Q_{all-br}$  | MPa             | Shear/bearing allowable load of the joint (plate material)                                                      |
| $RF$          | -               | Reserve Factor, i.e. ratio between allowable load and applied load                                              |
| $R_{br}$      | -               | Ratio between applied shear and shear-bearing allowable load                                                    |
| $R_{po}$      | -               | Ratio between applied tension and tension-pull-through allowable load                                           |
| $RF_{br}$     | -               | Ratio between bearing allowable load (not truncated by bolt shear strength) and bearing load                    |
| $t$           | mm              | Thickness of the plate                                                                                          |
| $T$           | N               | Tension load                                                                                                    |
| $T_{all-po}$  | N               | Tension pull-out/pull/through allowable load of the joint (plate material)                                      |
| $w$           | mm              | Width of the plate                                                                                              |
| $\sigma$      | MPa             | Axial stress                                                                                                    |
| $\tau$        | MPa             | Shear stress                                                                                                    |

### REFERENCES

|        | <i>Document Reference</i> | <i>Issue</i>        | <i>Title</i>                                                                                              |
|--------|---------------------------|---------------------|-----------------------------------------------------------------------------------------------------------|
| Ref. 1 | HSB21020-01               | Is. D               | Procedure for the determination of the static joint strength values of mechanically fastened metal joints |
| Ref. 2 | [SAE] MMPDS               | 9                   | Metallic Materials Properties Development and Standardization                                             |
| Ref. 3 | [M. C. Y. Niu]            | 2 <sup>nd</sup> Ed. | AIRFRAME STRESS ANALYSIS AND SIZING                                                                       |

### PLATE FAILURE: BEARING AND PULL-THROUGH

#### SOFTWARE

| <i>Program</i> | <i>Version</i> | <i>Description</i> |
|----------------|----------------|--------------------|
|                |                |                    |

Table 1. *Software programs referenced in this chapter*

#### RECORD OF REVISIONS

| <i>Revision Date</i> | <i>Authors</i> | <i>Change Reason</i>                                                                       | <i>Pages/Sub-chapters affected</i> |
|----------------------|----------------|--------------------------------------------------------------------------------------------|------------------------------------|
| 1.0<br>26/06/2019    | G. Baños       | First issue                                                                                | All pages                          |
| 1.1<br>04/03/2020    | G. Moreda      | Tables numbering corrected and Approval Signatures table updated according to new template | Pages 5 & 6                        |
| 1.2<br>04/12/2020    | G. Baños       | Added 0.6 correction factor to pull-through tension allowable per Niu's book chapter 9.9   | Pages 2 & 4                        |

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

#### APPROVAL SIGNATURES TABLE

| <i>Dpt. \ Site</i>                | <i>Getafe</i>          | <i>Manching</i>             |
|-----------------------------------|------------------------|-----------------------------|
| <b>Stress Methods (TEPAP-TL4)</b> | G. Baños<br>04/12/2020 | Florian Glock<br>04/12/2020 |

Table 3. *Approval signatures table for last revision of this chapter*